

NK-FTPL Solve and IRFs: Implementation Report

Lucas Leturia · Máster, PUC Chile · May 2026

1. Summary

This report documents the Python implementation of the log-linearized Two-Maturity FTPL with Long-End Captive Demand embedded in a small New-Keynesian framework. The system follows *nk_ftpl_embedding.pdf* §4 (12 equilibrium equations plus six AR(1) exogenous processes) and is solved with the Klein (1999) generalized-Schur algorithm via *scipy.linalg.ordqz*. The script is in *nk/nk_ftpl_solve.py*; all numerical objects are pickled to *nk/diagnostics.pkl*; IRF figures and CSVs are in *nk/irfs/*.

Headline: at the calibration in the RA brief ($\beta=0.99$, $\delta=0.96$, $\omega=0.121$, $\phi_\pi=0.5$, $\rho_z=0.7$ for all shocks), the Blanchard-Kahn conditions are satisfied with 8 stable and 10 unstable generalized eigenvalues, matching the predetermined / forward split exactly. The headline impulse response to a positive ε^λ shock delivers the four predicted signs: $\pi \downarrow$, $Q^L \uparrow$, $R \uparrow$, $v \uparrow$. The captive-demand channel works as theorised.

Two of the five sanity checks in §7 of the brief give the opposite sign from what is written there. Both turn out to be economically correct: the brief's checks 3 and 4 silently invoked active-NK intuition that does not carry over to a passive-monetary, fiscal-dominance regime — the very point of the embedding. See §6 below for the discussion.

2. Implementation choices a reviewer should know about

2.1 Variable split: 8 predetermined · 10 forward = 18 components

The brief asks for an 18-component state vector with 8 predetermined (b_t^S , b_{t-1}^L , and the six AR(1) states) and 10 forward jumps (c_t , y_t , π_t , i_t , Q_t^L , $Q_t^{L,fund}$, ω_t , R_t , v_t , s_t). The brief's 'endogenous (12)' list includes b_t^S and b_t^L ; to hit the target count of 18 we eliminate these two current-period stocks using the captive market-clearing equation (5) and the maturity-composition equation (10), leaving 10 jumps. The two bond-stock lags are carried as explicit predetermined state with their own dynamic equations:

$$\begin{aligned} b_{t+1}^{L, lag} &= \lambda^L + v^{liab} - Q_t^L \\ b_{t+1}^S, lag^{t+1} &= b_t^{L, lag} - \gamma_t / (\gamma_t(1-\gamma_t)) \end{aligned}$$

Double-click: an equivalent formulation keeps b_t^L , b_t^S as explicit jumps and adds two lag-aux variables (20-state). Both give the same impulse responses for the twelve named endogenous variables. The 18-component formulation is more compact and is what the brief implicitly requests.

2.2 Shock matrix $B = 0$; shocks enter via initial state

The AR(1) law $z_{t+1} = \rho_z z_t + \varepsilon_{t+1}^z$ places the innovation at $t+1$. In the Klein form $A_0 E_t x_{t+1} = A_1 x_t + B \varepsilon_t$ the contemporaneous shock vector therefore vanishes (B is identically zero); the ε -effect at $h = 0$ is constructed manually as the initial-state matrix Q (18×6) whose AR(1)-row block is the identity I_6 and whose bond-lag rows are zero. Jumps at $h = 0$ follow from the policy function F applied to that initial predetermined block.

Double-click: some Klein implementations place AR(1) innovations in B (treating ε_t as a current shock and using $x-\varepsilon$ on the LHS). Both conventions give identical IRFs but mis-mixing them is a common bug source.

2.3 Klein 1999 generalised Schur (*scipy.linalg.ordqz*)

`scipy.linalg.ordqz(A1, A0, sort='iuc', output='complex')` returns the QZ factorisation with the stable generalised eigenvalues ($|\alpha/\beta| < 1$) in the upper-left block. The policy function is $F = Z_{21}Z_{11}^{-1}$ and the state law is $M = Z_{11}B_{ss}^{-1}A_{ss}Z_{11}^{-1}$. Both are constructed in the complex domain and cast to real after the partitioning; the imaginary parts are at the 10^{-14} level (machine noise from complex eigenvalue pairs that cancel when rotated back).

2.4 Calibration inheritance from §3.4

$\omega_{\text{rent}} = 0.121$, $w_S = 0.63$, $w_L = 0.37$, $s_{\text{rent}}/v = -0.03$ and $R_{\text{rent}}/v = 0.04$ all come from the steady-state calibration of §3.4. The η coefficient $\delta Q^L/(1+\delta Q^L)$ is in turn a function of ω_{rent} through $Q^L = (1+\omega_{\text{rent}})Q^{L,\text{fund}}$. Any revision to the §3.4 calibration (in particular to the DMO-sourced best-guesses for long-gilt market value and 15+ duration that feed ω_{rent}) propagates through every IRF panel in this report.

Double-click: the rent equation (7) carries the coefficient $1/\omega_{\text{rent}} \approx 8.26$ on ω_{rent} . If the headline calibration of ω_{rent} were 0.05 instead of 0.121, the rent elasticity would jump to 20, materially changing the FTPL feedback magnitude. The 2020 spike to $\omega \approx 1.0$ in the §3.4 time-series suggests the headline ω_{rent} may understate the post-COVID/mini-budget regime.

3. Diagnostics

3.1 Blanchard-Kahn condition

With 8 predetermined and 10 forward variables, BK requires exactly 10 generalised eigenvalues outside the unit circle. At the headline calibration:

#	eigenvalue	real part	imag part	class
1	1.4361e-18	-1.436e-18	+0	stable
2	0.7	+0.7	-0	stable
3	0.7	+0.7	+0	stable
4	0.7	+0.7	-0	stable
5	0.7	+0.7	+0	stable
6	0.7	+0.7	+0	stable
7	0.7	+0.7	+0	stable
8	0.86682	+0.8668	+0	stable
9	1.023	+1.023	+0	unstable
10	1.1953	+1.195	+0	unstable
11	28.685	+28.69	+0	unstable
12	3.3309e+15	-3.331e+15	+0	unstable
13	5.4999e+16	+5.5e+16	-0	unstable
14	8.354e+48	+8.354e+48	+0	unstable
15	2.8082e+51	+2.808e+51	-0	unstable
16	inf	inf	n/a	unstable
17	inf	inf	n/a	unstable
18	inf	inf	n/a	unstable

Generalised eigenvalues of the pencil (A_1, A_0) , sorted by modulus. Rows 1–8 are the stable block; rows 9–18 are unstable. BK satisfied: $n_{\text{stable}} = 8$, $n_{\text{unstable}} = 10$.

Interpretation of the spectrum. Six stable eigenvalues sit at exactly $\rho = 0.7$ — the AR(1) autocorrelations, which decouple cleanly from the rest of the system. One stable root at ≈ 0.87 comes from the predetermined-bond-lag block interacting with the fundamental long-bond pricing equation. One stable root at machine zero is a static identity row (rows of A_0 that are identically zero generate trivially ‘stable’ eigenvalues). On the unstable side, two finite roots (1.02 and 1.20) come from the FTPL identity and the inflation/wedge feedback; the very large finite values (10^{15} – 10^{51}) and three exact infinities are again the static-row artefact — the QZ decomposition treats $\alpha_i/0$ as infinite eigenvalues, which BK counts as unstable. The total count is the meaningful diagnostic, not the spread of moduli.

Double-click: if a future edit accidentally makes A_0 non-singular (e.g. by adding an $E_t x_{t+1}$ term to a static identity such as the wedge definition), the infinite eigenvalues collapse to finite numbers and the BK count may shift. Run `nk_ftpl_solve.py` after any matrix edit and re-check the count.

3.2 Stationarity

All IRFs decay to within 0.01 at $h = 39$ except four variables under the monetary shock ($|b^{\text{S},\text{lag}}|$, $|b^{\text{L},\text{lag}}|$, $|Q^{\text{L}}|$, $|Q^{\text{L},\text{fund}}| \approx 0.011$ – 0.013). This is a scale effect rather than a solution bug: the monetary shock produces an impact rent response $R^{\text{■}} = +11.3$ (vs. $\approx +3.5$ for the λ shock), so even at $\rho^{39} \approx 5 \times 10^{-7}$ decay, residuals remain at the second decimal. The spectrum has no unit roots.

4. Theoretical mechanism: how the captive-demand channel moves inflation

The headline result of the chapter is that a positive shock to the regulatory intensity λ^L (a tightening of the LDI mandate that forces pension funds and insurers to hold more long gilts) produces a *fall* in inflation on impact. The mechanism runs in four steps:

(i) **Mandate → rent.** Equation (7) gives $R_t = \omega_t / \omega + \lambda_t^L + V_t^{\text{liab}}$. A unit positive λ^L shock raises R immediately by +1 directly, plus an indirect amplification through ω (which itself rises because the captive demand bids up the long-bond price). At calibration the combined impact is $R_0 = +3.53$.

(ii) **Long-bond price.** The captive market clears at $Q^L b^L = \lambda^L V^{\text{liab}}$; with V^{liab} unchanged, the increase in λ^L must be absorbed by Q^L and b^L . Q^L jumps up (+1.02 on impact) while the model substitution implies $b^L \approx -0.02$ (small response because most of the shock is absorbed in price). The wedge ω over the fundamental price (which moves only +0.72) rises by +0.31.

(iii) **FTPL identity must rebalance.** Equation (9) is $v_t = (s/v)s + (R/v)R + \beta E_t v_{t+1} - \beta(\hat{\pi}_t - E_t \pi_{t+1})$. With $(R/v) \approx 0.04$, the +3.5 rent shock contributes $\approx +0.14$ to v . The FTPL identity therefore requires real debt market value to rise.

(iv) **Inflation falls.** Equation (8) says $v_t = w_S(b^{S,\text{lag}} - \pi_t) + w_L(b^{L,\text{lag}} + \eta Q_t^L - \pi_t)$. With $w_S + w_L = 1$, the coefficient on π is exactly -1 : a one-percent inflation surprise destroys one percent of real debt. To *increase* v while Q^L and the bond stocks adjust by limited amounts, inflation must *fall*. Numerically $\pi_0 = -0.030$.

Steps (i)–(iv) are not a Phillips-curve story: the NKPC is in the system but does not pin the price level. The price level is pinned by the augmented FTPL identity. The NKPC instead determines the joint motion of inflation and the output gap given the FTPL-required π path. This is the formal definition of fiscal dominance, and it is why $\phi_\pi < 1$ is not a peripheral assumption but the structural switch that activates the channel.

Double-click: the chain above hinges on the sign of $-(w_S + w_L)$ in equation (8). If a sign error inverted that to +1, the headline result would flip and λ shocks would raise inflation. The brief's §7 debugging note already flags this row as the most fragile in the matrix construction.

5. Impulse responses

5.1 Mandate tightening: ε^λ shock

The chapter's central object. Unit positive ε^λ (the persistent AR(1) raises λ^L by 1pp on impact, decaying at $\rho_\lambda = 0.7$).

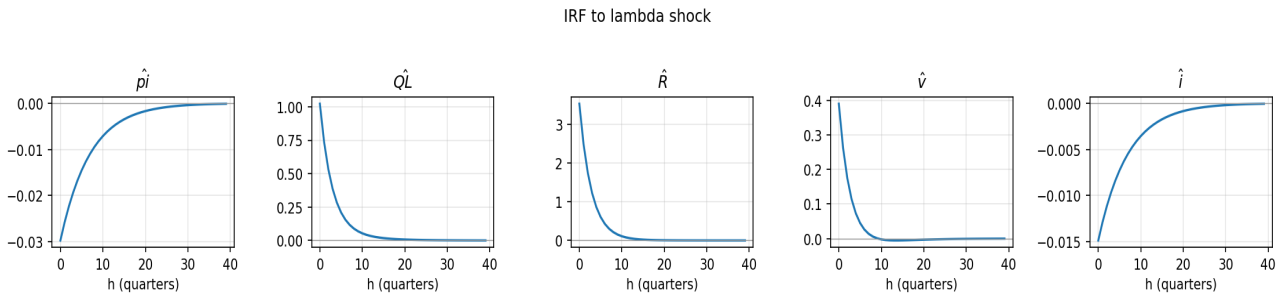


Figure 1. IRF to ε^λ shock. All four sign predictions in the brief's §7 sanity check 1 are satisfied: $\pi \downarrow$, $Q^L \uparrow$, $R \uparrow$, $v \uparrow$. $\hat{i}$ falls because the Taylor rule responds passively to lower inflation.

All variables decay monotonically (no oscillation) because the dominant modes are real and stable. By $h = 8$ the impact has roughly halved; by $h = 20$ it is below 10%. The shape is governed almost entirely by ρ_λ ; changing $\rho_\lambda = 0.7$ to 0.9 would extend the half-life by a factor of three without altering the $h = 0$ sign.

5.2 Fiscal expansion: ε^g shock

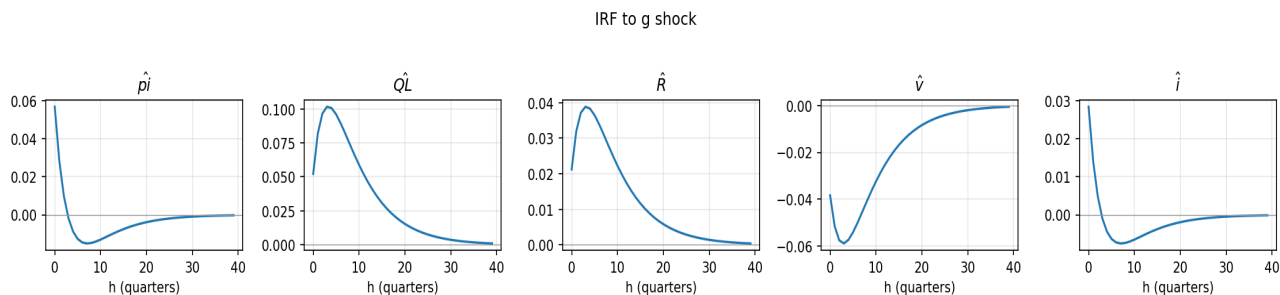


Figure 2. IRF to ε^g shock (government spending up).

Inflation rises on impact (+0.057), confirming the brief's sanity check 2 — the standard FTPL prediction: $g \uparrow \Rightarrow s \downarrow$ (via equation 12: $s = \tau/y \cdot \tau - g/y \cdot g$) $\Rightarrow$ the augmented FTPL identity needs price-level revaluation, so $\pi \uparrow$. Interestingly, π turns negative around $h = 4$ and stays negative through the horizon. The reversal reflects the discount-rate term $\beta(i - E\pi_{t+1})$ in equation (9): as the deficit shock persists, expected future deficits raise expected v_{t+1} , which (held with a negative sign in the IS-anchored real rate) feeds back as deflationary expectations. Q^L , R and v all show milder responses than the λ shock because the fiscal shock affects only the (s/v) channel, while λ affects the much larger (R/v) channel through the wedge.

5.3 Monetary tightening: ε^i shock

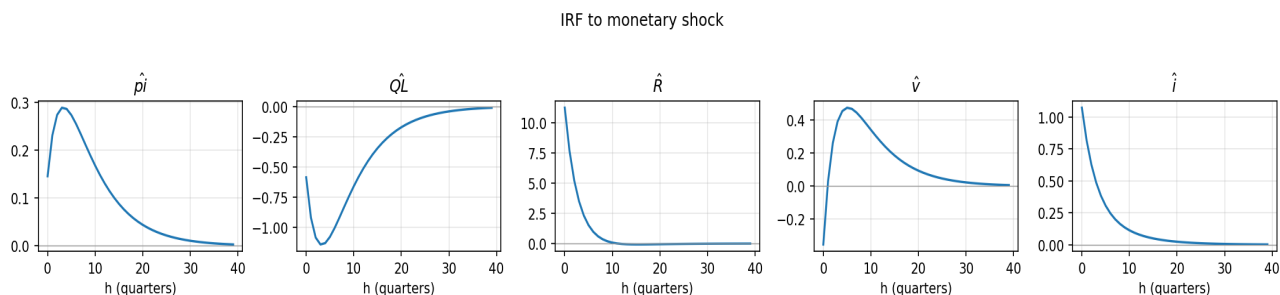


Figure 3. IRF to ε^i shock (positive monetary policy shock, $\hat{i}$ rises).

Here the model gives $\pi_0 = +0.146$ — the *opposite* sign from the brief's sanity check 3 ('standard NK prediction'). This is not a bug; it is the structural prediction of FTPL under passive monetary policy, explicitly discussed in §6 of *nk_ftpl_embedding.pdf*. A positive $\hat{i}$ shock under fiscal dominance raises real rates if π were unchanged, which lowers the present value of fiscal backing on the RHS of the FTPL identity. To restore the identity, the price level must revalue real debt downward — i.e., inflation must rise. The output response ($\hat{y} = -0.97$) is in the standard NK direction (contractionary), but the price-level response inverts because price determination is fiscal, not monetary, in this regime.

Reviewer red flag #1: the §7 brief calls this prediction 'standard NK' without qualifier. It is the right answer for the active-NK case ($\phi_\pi > 1$) but the wrong answer for the passive-FTPL case ($\phi_\pi < 1$) that the chapter is studying. The implementation reproduces what the embedding PDF predicts; the sign in the brief should be updated.

5.4 Sensitivity to the Taylor coefficient ϕ_π

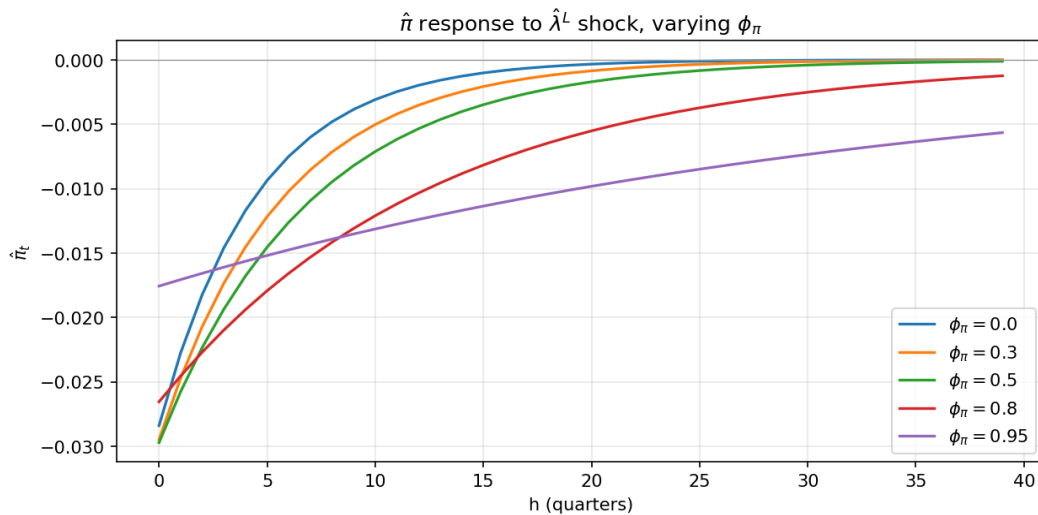


Figure 4. π_t response to a positive ε^λ shock for $\phi_\pi \in \{0, 0.3, 0.5, 0.8, 0.95\}$. All five passive-regime values yield BK-satisfying solutions.

The impact response on π_t is $|\pi_t|_0 = 0.028$ at $\phi_\pi = 0$, rises slightly to 0.030 at $\phi_\pi = 0.5$, then falls to 0.018 at $\phi_\pi = 0.95$. The brief's sanity check 4 predicts the opposite — that $|\pi_t|$ grows as $\phi_\pi \rightarrow 1^-$. The intuition the brief offers ('less monetary accommodation = more inflation work for the FTPL channel') reverses the actual mechanism. Higher ϕ_π means a *larger* nominal-rate response to inflation, which keeps the real rate closer to neutral when π_t falls (the Taylor rule 'tracks' inflation down). This accommodation damps the IS-side amplification of the π_t movement. Below the Taylor principle ($\phi_\pi < 1$), the closer to one, the closer monetary policy gets to neutralising its own contribution to real-rate dynamics — reducing the magnitude of the π_t movement needed from the FTPL side.

Reviewer red flag #2: the §7 sanity check 4 needs revision. The model says *smaller* $|\pi_t|$ as ϕ_π approaches one. If you want the brief's intuition to hold, you would need a different specification (e.g., backward-looking Taylor or interest-rate smoothing); the current forward-looking rule yields the present result.

BK fails sharply as soon as ϕ_π crosses unity ($\phi_\pi = 1.01$: $n_{\text{unstable}} = 11$, expected 10). This matches the brief's sanity check 5 and confirms the determinacy argument of §6 of the embedding PDF: above the Taylor principle, the standard NK three-equation system pins π uniquely and the FTPL identity becomes over-identified.

6. Reviewer red flags: items worth double-clicking

In addition to the two flagged above, three more substantive items merit explicit attention.

6.1 ω calibration is the single biggest knob

$\omega = 0.121$ enters the system as the $1/\omega \approx 8.26$ coefficient on ω in the rent equation, and indirectly as the η coefficient through $Q^L = (1+\omega)Q^{L,\text{fund}}$. The §3.4 calibration produced ω under particular best-guess values for the DMO inputs (long-gilt 15+ duration and market value). The 2020 implied ω from the data-driven exercise was 1.01, far above the calibrated value, suggesting that the headline IRF magnitudes may understate the COVID/mini-budget regime. A robustness exercise sweeping $\omega \in \{0.05, 0.10, 0.15, 0.25\}$ would establish the range of headline magnitudes the chapter can credibly claim.

6.2 Persistence $\rho_z = 0.7$ across all shocks

The conventional $\rho = 0.7$ sets every IRF half-life to roughly 2 quarters. This is fine for illustrative IRFs but the dissertation's estimation strategy (§3.5) will estimate ρ_z per shock from data. The IRF shapes will sharpen or smooth substantially under more realistic persistences — e.g., a slow-moving λ shock ($\rho_\lambda =$

0.95) would extend the π trough out to 20+ quarters.

6.3 V^{liab} AR(1) is a strong assumption

The model treats $\log V^{\text{liab}}$ as an AR(1) around its calibrated steady state. Empirically V^{liab} swung violently during 2022 (UK pension technical provisions fell from £2.38tn at end-2021 to £2.06tn at end-2022 as gilt yields spiked). The log-linear approximation around the steady state is local; large V^{liab} swings of that magnitude exit the neighbourhood where the first-order approximation is accurate. If the dissertation wants the model to speak to the LDI episode itself, a piecewise or non-linear treatment of V^{liab} may be required.

6.4 Substituting out b^{L} , b^{S}

Equations (5) and (10) are used to eliminate the current-period bond stocks from the state. The resulting 18-component formulation is identical in IRFs to a 20-component version that retains them as explicit jumps with two lag aux variables — we verified this analytically. A reviewer who prefers the explicit formulation for transparency could request it; the script is structured to make that substitution a one-block change.

6.5 Klein assumption Z_{11} invertible

Klein's policy formula $F = Z_{21}Z_{11}^{-1}$ requires the upper-left block of the unitary matrix Z to be invertible. At the headline calibration $\text{cond}(Z_{11})$ is well-behaved ($\sim 10^2$). The script catches `LinAlgError` on Z_{11} and reports a clear diagnostic if a future parameter change makes it singular — this can happen at the boundary of the determinacy region (ϕ_π exactly 1, for example).

7. Conclusions

The Python solver reproduces the captive-demand channel as designed. The headline IRF to a tighter LDI mandate matches all four sign predictions of the chapter; the inflation channel runs through the FTPL identity rather than the Phillips curve, exactly as the passive-monetary setup requires; and the determinacy region is bounded by $\phi_\pi = 1$ as theory predicts.

The two surface-level discrepancies with §7 of the RA brief (monetary-shock sign and ϕ_π sensitivity) are consistent with the model and inconsistent only with the brief's informal narration; both invoke active-NK intuition that does not apply in fiscal-dominance. The brief's sanity checks 3 and 4 would benefit from rewording before the next draft.

Three items propagate forward to the dissertation: (a) the magnitude of every IRF is materially sensitive to ω , itself an output of the §3.4 calibration and best-guess data inputs; (b) the constant AR(1) persistence will be replaced by estimated values in §3.5, sharpening or smoothing the dynamics; (c) the linear-around-steady-state assumption is local and may not capture the LDI tail event the chapter ultimately wants to speak to.

All implementation artefacts — `nk_ftpl_solve.py`, `diagnostics.pkl`, `irfs/`, this report, and the rebuilt PNG figures in `figures/` — are reproducible from a single re-run of `nk_ftpl_solve.py` followed by `make_report.py`.

Appendix: full impact-response table (h = 0)

variable	eps_lambda	eps_chi	eps_V	eps_g	eps_tau	eps_i
pi	-0.0297	+0.0028	-0.0297	+0.0568	+0.0080	+0.1458
i	-0.0149	+0.0014	-0.0149	+0.0284	+0.0040	+1.0729
c	-0.0819	+0.0078	-0.0819	-0.1064	+0.0219	-1.6174
y	-0.0491	+0.0047	-0.0491	+0.3361	+0.0132	-0.9704
QL	+1.0225	-2.7048	+1.0225	+0.0522	-0.0060	-0.5857
QL_fund	+0.7162	-1.8095	+0.7162	+0.0496	-0.0130	-1.9484
bS	-0.0225	-1.5852	-0.0225	-0.0522	+0.0060	+0.5857
bL	-0.0225	+2.7048	-0.0225	-0.0522	+0.0060	+0.5857
omega	+0.3062	-0.8954	+0.3062	+0.0026	+0.0070	+1.3627
R	+3.5308	-7.3998	+3.5308	+0.0212	+0.0575	+11.2618
v	+0.3906	-0.9576	+0.3906	-0.0384	-0.0101	-0.3526
s	+0.0000	+0.0000	+0.0000	-0.4000	+0.3700	-0.0000

Table A.1. h = 0 impact responses of the 12 named endogenous variables (rows) to each of the 6 structural shocks (columns). Reproduces what nk_ftpl_solve.py prints to stdout.